

---

# NUMERICAL TECHNIQUES AND DIFFERENTIAL EQUATIONS

---

**Course Code: MAC21**

**Credits: 2:1:1**

**Pre-requisites: -**

**Contact Hours: 28L+14T+14P**

**Course Coordinators: Dr. Monica Anand & Dr. B. Azghar Pasha**

---

## Course Content

### Unit I

**Series Expansion of Functions & their Applications:** Taylor's series and Maclaurin's series expansion of one variable (without proof) and its applications to solve algebraic and transcendental equations - Newton-Raphson method.

Taylor's series and Maclaurin's series expansion of two variables (without proof) and its applications to solve the system of nonlinear equations - Newton-Raphson method. Maxima and minima of functions of two variables, Lagrange's method of undetermined multipliers.

- Pedagogy/Course delivery tools: Chalk and talk, Power Point Presentation, Videos
- Links: <https://nptel.ac.in/courses/111/106/111106101/>  
<https://nptel.ac.in/courses/111/105/111105121/>
- Impartus recording: <https://a.impartus.com/ilc/#/course/119640/593>  
<https://a.impartus.com/ilc/#/course/59742/295>  
<https://a.impartus.com/ilc/#/course/619570/1030>

### Unit II

**First order and first degree differential equations:** Applications of first order and first degree differential equations to solve simple electric circuit problems, Newton's law of cooling and orthogonal trajectories.

**Numerical solution of first order ODE:** Taylor's series method, Euler's & modified Euler's method, fourth order Runge-Kutta method.

- Pedagogy/Course delivery tools: Chalk and talk, Power Point Presentation, Videos
- Links: <https://nptel.ac.in/courses/111/106/111106100/>  
<https://nptel.ac.in/courses/111/106/111106101/>
- Impartus recording: <https://a.impartus.com/ilc/#/course/119640/593>  
<https://a.impartus.com/ilc/#/course/59742/295>  
<https://a.impartus.com/ilc/#/course/619570/1030>

### Unit III

**Linear differential equations of higher Order:** Linear differential equations of higher order with constant coefficients. Cauchy's and Legendre's linear differential equations, Method of variation of parameters – Engineering applications.

- Pedagogy/Course delivery tools: Chalk and talk, Power Point Presentation, Videos
- Links: <https://nptel.ac.in/courses/111/105/111105121/>
- Impartus recording: <https://a.impartus.com/ilc/#/course/119640/593>  
<https://a.impartus.com/ilc/#/course/59742/295>

## Unit IV

**Interpolation:** Forward and Backward differences, Interpolation, Newton-Gregory Forward and Backward Interpolation, Lagrange's interpolation and Newton's divided difference interpolation techniques (no proof).

**Numerical Differentiation:** Derivatives using Newton-Gregory forward and backward interpolation formula.

**Numerical Integration:** Newton-Cotes quadrature formula: Trapezoidal, Simpson's 1/3rd and Simpson's 3/8th rule (no proof).

- Pedagogy/Course delivery tools: Chalk and talk, Power Point Presentation, Videos
- Links: <https://nptel.ac.in/courses/111/106/111106101/>
- Impartus recording: <https://a.impartus.com/ilc/#/course/96127/452>  
<https://a.impartus.com/ilc/#/course/132243/636>

## Unit V

**Linear Algebra:** Elementary transformations on a matrix, Echelon form & rank of a matrix, Consistency of system of linear equations. Gauss elimination, Gauss – Seidel method to solve system of linear equations. Eigen values and eigen vectors of a matrix, Rayleigh power method to determine the dominant eigen value of a matrix, diagonalization of square matrices. Solution of system of ODEs by matrix method.

- Pedagogy/Course delivery tools: Chalk and talk, Power Point Presentation, Videos
- Links: <https://nptel.ac.in/courses/111/108/111108066/>  
<https://nptel.ac.in/courses/111/105/111105035/>
- Impartus recording: <https://a.impartus.com/ilc/#/course/619570/1030>

### Text Book:

1. **George B. Thomas, Maurice D. Weir, Joel R. Hass** - Thomas' Calculus, Pearson, 13<sup>th</sup> edition, 2014.
2. **B.S. Grewal** – Higher Engineering Mathematics, Khanna Publishers, 44<sup>th</sup> edition, 2017.

### Reference Books:

1. **Erwin Kreyszig** - Advanced Engineering Mathematics, Wiley publication, 10<sup>th</sup> edition, 2015.
2. **Peter V. O' Neil** - Advanced Engineering Mathematics, Thomson Brooks/ Cole, 7<sup>th</sup> edition, 2011.
3. **Glyn James & Phil Dyke** - Advanced Modern Engineering Mathematics, Pearson Education, 5<sup>th</sup> edition, 2018.
4. **Srimanta Pal & Subobh C Bhunia** - Engineering Mathematics, Oxford University Press, 3<sup>rd</sup> Reprint, 2016.

## Course Outcomes (COs):

At the end of the course the student will be able to:

1. Determine extreme values, roots of algebraic & transcendental equations and solution of non-linear system of equations. (PO-1, PO-2)
2. Solve first order differential equations analytically and numerically. (PO-1, PO-2)
3. Solve higher order linear differential equations with constant and variable coefficients. (PO-1, PO-2)
4. Interpolate, differentiate and integrate a given set of tabulated data (PO-1, PO-2)
5. Solve system of linear equations and ordinary differential equations using matrices. (PO-1, PO-2)

## List of Programs:

1. Analyzing extreme values of functions of two variables
2. Finding the roots of algebraic and transcendental equations using Newton-Raphson method
3. Solution of system of non-linear equations using Newton-Raphson method
4. Taylor's series and Euler's method to solve ODE's of first order and first degree
5. Modified Euler's method & Runge-Kutta method of 4<sup>th</sup> order to solve first order and first degree ODE's
6. Solution of higher order ODE's
7. Interpolation/Extrapolation for equispaced and unequispaced data
8. Evaluation of definite integrals using Trapezoidal, Simpson's (1/3)<sup>rd</sup> and (3/8)<sup>th</sup> rule
9. Gauss-Seidel iteration method to solve the system of linear equations
10. Finding the largest eigen value and the corresponding eigen vector using Rayleigh power method.

## Course Assessment and Evaluation:

| <b>Continuous Internal Evaluation (CIE): 50 Marks</b>         |              |                                 |
|---------------------------------------------------------------|--------------|---------------------------------|
| <b>Assessment tool</b>                                        | <b>Marks</b> | <b>Course outcomes attained</b> |
| Internal Test-I                                               | 30           | CO1, CO2, CO3                   |
| Internal Test-II                                              | 30           | CO3, CO4, CO5                   |
| Average of the two tests shall be taken                       |              |                                 |
| <b>Lab components</b>                                         |              |                                 |
| Conducting Experiment and Laboratory Record                   | 10           | CO1, CO2, CO3, CO4, CO5         |
| Lab Test                                                      | 10           | CO1, CO2, CO3, CO4, CO5         |
| <b>Semester End Examination (SEE)</b><br>(Scaled to 50 Marks) | 100          | CO1, CO2, CO3, CO4, CO5         |

---

# ENGINEERING CHEMISTRY

---

**Course Code:** CYC22

**Credits:** 3:0:0

**Pre-requisites:** -

**Contact Hours:** 42L

**Course Coordinators:** Dr. Nagaraju Kottam & Dr. R Hari Krishna

---

## Course Content

### Unit I

#### Chemical Energy Conversion and Battery Technology

**Chemical Energy Conversion:** Basic concepts of electrochemistry – electrode potential, origin of single electrode potential, Galvanic cells. Derivation of Nernst equation, Types/ Reference electrodes – Calomel electrode: Construction and working, advantages and its applications. Ion selective electrodes (Glass electrode). Determination of pH of an unknown solution using glass electrode. Concentration cells. Numerical problems on electrode potential, EMF of cells and concentration cells.

**Battery Technology:** Basic concepts. Mechanism of battery operation, battery characteristics. Classification of batteries – Primary, secondary and reserve batteries. Modern batteries: construction, working and applications of Nickel-Metal hydride battery, Metal-air batteries (Ex: Zn-air battery) and Li-ion (Lithium batteries).

**Self-study:** New generation Batteries: Introduction and Example- Al-air batteries.

- Pedagogy/Course delivery tools: Chalk and talk, Power Point Presentation, Videos
- Lab component/Practical Topics: Potentiometric titration using calomel & platinum electrode assembly. Determination of electrode potential of unknown metal using Nernst equation
- Links: Impartus video lectures: <http://msrit.edu/study-material/chemistry.html>

### Unit II

#### Corrosion Science and Engineering

**Corrosion Science and Engineering:** Metallic corrosion - Definition, electrochemical theory of corrosion. Types of corrosion - Differential metal corrosion, differential aeration corrosion (Ex: pitting and waterline corrosion) and stress corrosion. Factors affecting the rate of corrosion.

**Corrosion Control Methods:** Chemical Methods: Inorganic coatings- anodizing and phosphating. Corrosion inhibitors. Physical Methods: Cathodic Protection-sacrificial anode method.

**Metal Coating:** Anodic and cathodic metal coatings with examples.

**Self-study:** Multifunctional coatings (Thermal resistance, photo resistance and scratch resistance).

- Pedagogy/Course delivery tools: Chalk and talk, Power Point Presentation, Videos
- Lab component/Practical Topics: Analysis of iron content in steel solution by external indicator method.
- Links: Impartus videos: <http://msrit.edu/study-material/chemistry.html>

## Unit III

### Polymer Science, Liquid Crystal and Display Devices

**Polymer Science:** Introduction to high polymers. Glass transition temperature ( $T_g$ ), factors influencing  $T_g$ . Conducting polymers - Definition, mechanism of conduction in polyacetylene. High performance (Engineering) polymers: Synthesis, properties and applications of Teflon and PMMA. Polymer composites- Introduction. Synthesis and applications of carbon fiber.

**Liquid Crystal:** Introduction, meaning, positional and orientational order in solid, liquid crystals and liquids. Director. Classification – thermotropic and lyotropic with examples. Liquid crystalline behavior in homologues series – PAA series. Applications of liquid crystals. Applications of liquid crystals in seven segment display system.

**Display Devices:** Properties and applications of Organic Light Emitting Diodes (OLED's) and Quantum Light Emitting Diodes (QLED's)

**Self-study:** Types of mesophases – Nematic, Chiral nematic (cholesteric), Smectic and Columnar – arrangements of molecules with examples.

- Pedagogy/Course delivery tools: Chalk and talk, Power Point Presentation, Videos
- Lab component/Practical Topics: Synthesis of urea-formaldehyde
- Links: Impartus videos: <http://msrit.edu/study-material/chemistry.html>

## Unit IV

### Nanomaterials and Electrochemical Sensors

**Nanomaterials:** Introduction to nanomaterials. Synthesis: top-down and bottom-up approaches. Chemical methods of synthesis- solution combustion and hydrothermal methods. Characterization techniques like PXRD, SEM, and TEM (only mention & introduction). Applications of nanomaterials.

**Electrochemical Sensors:** Introduction to sensors. Principle and instrumentation of: potentiometric sensors and its application in the estimation of iron. Optical sensors (colorimetric) and its application in the estimation of the copper. Conductometric sensors and its application in the determination of acid mixture ( $\text{HCl}$  &  $\text{CH}_3\text{COOH}$ ) by conductometric method. Problems on Beer-Lamberts law.

**Self-study:** Technological applications of nano oxide materials.

- Pedagogy/Course delivery tools: Chalk and talk, Power Point Presentation, Videos
- Lab component/Practical Topics: Synthesis of nanomaterials by wet chemical methods, Determination of pH, Estimation of metals by colorimeter.
- Links: Impartus videos: <http://msrit.edu/study-material/chemistry.html>

## Unit V

### Fuel Cells, Solar Energy and Hydrogen Energy

**Fuel Cells:** Introduction, construction, working and applications of methanol–oxygen and polymer electrolyte fuel cell.

**Solar Energy:** Introduction, importance of solar PV cell, construction and working solar PV cell, advantages and disadvantages.

**Hydrogen Energy:** Introduction to Hydrogen as an Energy Source. Hydrogen Production methods: Hydrogen Production by Steam Methane Reforming, hydrogen production using electrochemical and photoelectrochemical water splitting. Hydrogen storage: different materials for storage – metal hydrides, high surface area materials, complex and chemical hydrides. Advantages and Disadvantages of hydrogen as an energy source.

**Self-study:** Electrodes for electrostatic double layer capacitors, pseudo capacitors and hybrid capacitor.

- Pedagogy/Course delivery tools: Chalk and talk, Power Point Presentation, Videos
- Links: Impartus videos: <http://msrit.edu/study-material/chemistry.html>

### Suggested Learning Resources:

#### Text Book:

1. **P. C. Jain and Monica Jain** - A text Book of Engineering Chemistry, Dhanapat Rai Publications, New Delhi, 17<sup>th</sup> edition, 2018.
2. **R.V. Gadag and Nithyananda Shetty** - A text Book of Engineering Chemistry, Medtech Publishers, 1<sup>st</sup> edition 2019.

#### Reference Books:

1. **F.W. Billmeyer** - Text Book of Polymer Science, John Wiley & Sons, 4<sup>th</sup> edition, 2007.
2. **M.G. Fontana, N. D. Greene** - Corrosion Engineering, McGraw Hill Publications, New York, 3<sup>rd</sup> edition, 2005.
3. **B.R. Puri, L.R. Sharma & M.S. Pathania**, - Principles of Physical Chemistry, S Nagin Chand & Co., 48<sup>th</sup> edition, 2019.
4. **G.A.Ozin and A.C. Arsenault** - Nanochemistry: A Chemical approach to Nanomaterials, RSC Publishing, 2005.
5. **S. Chandrashekar** - Liquid Crystals, Cambridge University Press, 2<sup>nd</sup> edition, 2010.
6. **Peter J. Collings** - Introduction to Liquid Crystals, CRC Press, 2<sup>nd</sup> edition, 2019.
7. **Dmitri Bessarabov, Haijiang Wang, Hui Li, Nana Zhao** - PEM Electrolysis for Hydrogen Production, CRC press, 1st edition, 2017.
8. **Santhanam K S V** - Introduction to hydrogen technology, 2<sup>nd</sup> edition, John-wiley and Sons Inc, 2018.

### Web links and Video Lectures (e-Resources):

1. <http://msrit.edu/study-material/chemistry.html>
2. <https://nptel.ac.in/courses/122/101/122101001/>

### Course Outcomes (COs):

At the end of the course the student will be able to:

1. Apply the knowledge of electrochemistry to improve the efficiency of batteries. (PO-1, PO-2, PO-7)
2. Interpret the reasons of corrosion, monitor and control by using the proper techniques. (PO-1, PO-2, PO-7)
3. Apply the knowledge in synthesis of advanced polymers, conducting polymers and liquid crystal materials for different applications. (PO-1, PO-2, PO-7)
4. Understand the synthesis of nanomaterials, principle and instrumentation of electrochemical sensors, and their applications. (PO-1, PO-2, PO-7)
5. Understand the working principle of different types of fuel cells, PV cell and hydrogen energy. (PO-1, PO-2, PO-7)

### Course Assessment and Evaluation:

| <b>Continuous Internal Evaluation (CIE): 50 Marks</b>         |              |                                 |
|---------------------------------------------------------------|--------------|---------------------------------|
| <b>Assessment tool</b>                                        | <b>Marks</b> | <b>Course outcomes attained</b> |
| Internal Test-I                                               | 30           | CO1, CO2, CO3                   |
| Internal Test-II                                              | 30           | CO3, CO4, CO5                   |
| Average of the two internal tests will be taken for 30 marks. |              |                                 |
| <b>Other components:</b>                                      |              |                                 |
| Assignment                                                    | 10           | CO1,CO2,CO3,CO4,CO5             |
| Quiz/MCQ                                                      | 10           | CO1,CO2,CO3,CO4,CO5             |
| Assignment and quiz/ MCQ = 20 marks.                          |              |                                 |
| <b>Semester End Examination (SEE)</b><br>(Scaled to 50 Marks) | 100          | CO1,CO2,CO3,CO4,CO5             |

---

## PROFESSIONAL WRITING SKILLS IN ENGLISH

---

**Course Code:** HSCC15/25

**Credits:** 1:0:0

**Pre-requisites:** Nil

**Contact Hours:** 14L+14P

**Course Coordinator:** Dr. Diwakar P

---

### Course Contents

#### Unit I

##### Identifying Common Errors in Writing and Speaking English

Identifying common errors while using the parts of speech, Use of verbs and phrasal verbs, Auxiliary verbs, Subject Verb Agreement (Concord rules), Identifying common errors in usage of Subject-verb agreement and tenses

- Pedagogy / Course delivery tools: Task based Teaching learning, Classroom sharing, Power point presentation, Videos.
- Lab component: English Language Lab – Orell English Language.
- Links: Communication skills: [www.bbcenglishlearning.com](http://www.bbcenglishlearning.com)

#### Unit II

##### Nature and Style of Sensible writing

Formal writing- Introduction, Principles & Structure of Formal writing, Importance of proper punctuation, Usage of connectives, Precise writing, Essay writing, words commonly confused/misused

- Pedagogy / Course delivery tools: Chalk and talk, Power point presentation, Videos.
- Lab component: English Language Lab – Orell English Language Software..

#### Unit III

##### Technical Writing

Introduction to technical writing, Introduction to Technical Proposal writing, Types of Technical Proposals, Characteristics of Technical Proposals, Grammar- Voices and Reported speech.

- Pedagogy / Course delivery tools: Task based Teaching learning, Classroom sharing, Power point presentation, Electric Approach
- Lab component: English Language Lab – Orell English Language Software – Grammar Exercises
- Links: Communication: <https://academicguides.waldenu.edu/writingcenter/grammar/subjectverbagreement>  
[www.bbcenglishlearning.com](http://www.bbcenglishlearning.com)



## Unit IV

### Professional Communication for Employment

Listening Comprehension, Types of Listening, Listening Barriers, Improving Listening Skills, Reading Comprehension, Techniques of Reading, Tips for effective reading, Job Application, Drafting effective resume/curriculum vitae, writing formal Email.

- Pedagogy / Course delivery tools: Task based Teaching learning, Classroom sharing, Power point presentation, Electric Approach
- Lab component / Practical Topics: English Language Lab – Orell English Language Software – Reading & Writing Exercise

## Unit V

### Professional Communication Skills at Workplace

Intra and Interpersonal communication skills at workplace, Significance of non-verbal communication Group Discussion and Presentation skills, strategies for effective GD, and Presentation skills, Interview skills.

- Pedagogy / Course delivery tools: Task based Teaching learning, Classroom sharing, Power point presentation.
- Lab component / Practical Topics: English Language Lab – Orell English Language Software – Presentation and GD Exercises.

[www.bbcenglishlearning.com](http://www.bbcenglishlearning.com)

### Text Books:

1. **Sanjay Kumar & Pushp Lata** - Communication Skills, Oxford University Press India Pvt Ltd- 2019.
2. **Meenakshi Raman & Sangeetha Sharma** - Technical communication – Principles and Practice, Oxford University Press, 2007.
3. **A.J. Thomson & A. V. Martinet** - A Practical English Grammar, Oxford University Press, 1987.

### Reference Books:

1. **M Ashraf Rizvi** - Effective Technical Communication, McGraw Hill Education (India) Private Limited-2005.
2. **Dr. Premila D Swamy & Udayakumar H M** - Communication skills for Engineers, Archers and Elevators Publishing House Bangalore, India, 2021.
3. **S K. Khandelwal & R K Gupta** - Functional Grammar & Composition, Laxmi Publication (P) Ltd.
4. **N. Krishna Murthy** - Modern English Grammar, Trinity press, 2016.
5. **John Seely** - The Oxford Guide to Effective Writing and Speaking. OUP, 2005.
6. Oxford Advanced Learners's Dictionary, 8<sup>th</sup> edition, 2013.
7. **Martin Hewings** - Advanced Grammar in Use, Cambridge University Press, 2013.

### Course Outcomes (COs):

At the end of the course the student will be able to:

1. Understand and identify the common errors in writing and speaking English. (PO-9, PO-10, PO-12)
2. Understand nuances of writing and enhance writing skills (PO-10, PO-12)
3. Learn and draft technical reports and writing. (PO-10, PO-12)
4. Use appropriate professional communication and writing practices. (PO-9, PO-10, PO-12)
5. Apply suitable presentation techniques in workplace environment. (PO-9, PO-10, PO-12)

### Course Assessment and Evaluation:

| <b>Continuous Internal Evaluation (CIE): 50 Marks</b>         |                                                          |                                  |
|---------------------------------------------------------------|----------------------------------------------------------|----------------------------------|
| <b>Assessment tool</b>                                        | <b>Marks</b>                                             | <b>Course outcomes addressed</b> |
| Internal test-I                                               | 30<br>(6 marks objective & 24 marks subjective pattern)  | CO1,CO2,CO3                      |
| Internal test-II                                              | 30<br>(6 marks objective & 24 marks subjective pattern)  | CO3,CO4,CO5                      |
| Average of the two internal tests will be taken for 30 marks. |                                                          |                                  |
| <b>Other components</b>                                       |                                                          |                                  |
| Assignment-I                                                  | 10                                                       | CO1, CO2, CO3                    |
| Assignment-II                                                 | 10                                                       | CO3, CO4, CO5                    |
| <b>Semester End Examination (SEE)</b>                         | 50<br>(10 marks objective & 40 marks subjective pattern) | CO1,CO2,CO3,CO4,CO5              |

---

## CONSTITUTION OF INDIA

---

**Course Code: HSCC16/26**

**Credits: 1:0:0**

**Pre-requisites: -**

**Contact Hours: 14L**

**Course Coordinator: Mrs. Kanya Kumari S**

---

### Course Contents

#### Unit I

##### **Introduction to the Constitution of India**

Making of the Indian Constitution and its salient features, Preamble of the Constitution, Fundamental Rights and relevant cases.

- Pedagogy/Course delivery tools: Chalk and Talk, power point presentation

#### Unit II

##### **Relevance of Directive Principles of State Policy -part-IV**

Fundamental Duties & their significance.

- Pedagogy/Course delivery tools: Chalk and Talk, power point presentation

#### Unit III

##### **Union executive and State executive**

The organs of government -Union executive- the President of India, Vice President, Prime Minister & Council of Ministers. The Union Legislature, Compositions & the functions of Parliament and the Supreme Court of India -composition & Jurisdictions State executive-Governor, Chief Minister & Council of Ministers, State legislature-composition & functions of legislative assembly & legislative council and State Judiciary.

- Pedagogy / Course delivery tools: Chalk and Talk, power point presentation

#### Unit IV

##### **Emergency provisions and special constitutional provisions in India.**

Emergency provisions, kinds effects and proclamation of emergencies

- Pedagogy / Course delivery tools: Chalk and Talk, power point presentation

#### Unit V

##### **Electoral process & major Constitutional Amendments**

Election commission of India & Electoral process, Amendment procedure and Major Constitutional amendments.

- Pedagogy / Course delivery tools: Chalk and Talk, power point presentation

## Text Books:

1. A Primer on Constitution of India & Professional ethics, VTU Publication-2007

## Reference Books:

1. **Durga Das Basu** - Introduction to Constitution of India - 19<sup>th</sup> / 20<sup>th</sup> edition 2001
2. **M.V.Pylee** - An Introduction to Constitution of India, 4<sup>th</sup> edition, 2008
3. **Dr. K. R. Phaneesh** - Constitution of India & Professional Ethics, Sudha publication, 10<sup>th</sup> revised edition 2018.

## Course Outcomes (COs):

At the end of the course the student will be able to:

1. Identify the fundamental principles of Indian Constitution.(PO-12)
2. Examine various provisions of the Directive principles of state policies and fundamental duties. (PO-6,PO-12)
3. Understand the powers & functions of executive, legislature and judicial system at the center and state level. ( PO-6, PO-12)
4. Identify the role of government. (PO-12)
5. Understand about process of election and amendment of Indian Constitution. (PO-6, PO-12)

## Course Assessment and Evaluation:

| <b>Continuous Internal Evaluation (CIE): 50 Marks</b>         |              |                                  |
|---------------------------------------------------------------|--------------|----------------------------------|
| <b>Assessment tool</b>                                        | <b>Marks</b> | <b>Course outcomes addressed</b> |
| Internal test-I                                               | 30           | CO1,CO2,CO3                      |
| Internal test-II                                              | 30           | CO3,CO4,CO5                      |
| Average of the two internal tests will be taken for 30 marks. |              |                                  |
| Other components                                              |              |                                  |
| Assignment                                                    | 10           | CO1, CO2                         |
| quiz                                                          | 10           | CO3, CO4, CO5                    |
| <b>Semester End Examination (SEE)</b>                         | <b>50</b>    | <b>CO1,CO2,CO3,CO4,CO5</b>       |

---

## DESIGN THINKING

---

**Course Code:** AECC17/27

**Credits:** 2:0:0

**Pre-requisites:** Nil

**Contact Hours:** 28L

**Course Coordinators:** Dr. Christina Grace, Dr.Prabha Ravi

---

### Course Content

#### Unit I

**Introduction:** Introduction to Design Thinking, Design and Business, Design Thinking for Education, Design Thinking Mindsets: Six Key Mindsets, Other Mindsets for Success

- Pedagogy/Course delivery tools: Chalk and Talk, Power point presentation, Videos
- Link: <https://www.youtube.com/watch?v=dAWwFG3X6u0>,
- <https://www.davidleedtech.org/>

#### Unit II

**The Design Thinking Process:** The Design Thinking Process, The Five Phases of Design Thinking: Empathize Phase, Self- Awareness and Partnerships, Interviews, Observations, Immersion, Research, Empathy Map, Projects, Define Phase, Synthesis: Finding Needs and Insights, Problem Statement, “How Might We” Question.

- Pedagogy/Course delivery tools: Chalk and Talk, Power point presentation, Videos
- Link: <https://www.youtube.com/watch?v=5CUt2QQsJfc>

#### Unit III

**Ideate Phase and Prototyping:** Ideate Phase, Demystifying Creativity, Innovation, and Originality, Ideate Principles, Pre- Brainstorming: Mindset, Warm-Ups, and Practice, Prototype Phase, Rapid Prototyping, Prototyping in Action, Facilitation and Mentorship, Makerspace, Tools, and Materials.

- Pedagogy/Course delivery tools: Chalk and Talk, Power point presentation, Videos
- Links: <https://youtu.be/M66ZU2PCicM>

#### Unit IV

**Testing Phase, Product life cycle and Value engineering:** Test Phase, Embracing Failure, Testing with End Users, Testing without End Users, Iteration; Product life cycle (Development, Introduction, Growth, Maturity, Decline), Value Engineering Job Plan, Case study on value engineering

- Pedagogy/Course delivery tools: Chalk and Talk, Power point presentation, Videos
- Links: <https://youtube.com/watch?v=qgVs8vskWl0&feature=shares>  
[https://youtube.com/watch?v=85\\_Eet4o9QM&feature=shares](https://youtube.com/watch?v=85_Eet4o9QM&feature=shares)  
<https://youtube.com/watch?v=CkezCE3GmeQ&feature=shares>

## Unit V

**Case studies:** Case studies from YouTube: Amul and its innovation during pandemic, Asian Paints growth strategy, Prototype of smart village, Futuristic farms, Data driven design,

Case study from Jeanne Liedtka et al book: Redesigning the Customer Contact Center at Toyota, Rethinking Subsidized Meals for the Elderly at The Good Kitchen

- Pedagogy/Course delivery tools: Chalk and Talk, Power point presentation
- Link: <https://youtu.be/nnwqtZiYMxQ>  
<https://youtu.be/jGT6ob8hV6M>  
<https://youtu.be/SlhE4--7IEM>  
<https://youtu.be/KfB2sx9uCkI>  
<https://youtu.be/Jh5xKBuvMIA>

### Text Book:

1. **David Lee**, Design Thinking in the Classroom, Ulysses Press, Korea, 2018

### Reference Books:

1. **Jeanne Liedtka, Andrew King, Kevin Bennett** - Solving Problems with Design Thinking - Ten Stories of What Works (Columbia Business School Publishing) Hardcover –2013
2. **Roger Martin** - The Design of Business: Why Design Thinking is the Next Competitive Advantage, Harvard Business Press, 2009.
3. **Idris Mootee** - Design Thinking for Strategic Innovation: What They Can't Teach You at Business or Design School, John Wiley & Sons 2013.

### Web links and Video Lectures (e-Resources):

1. Stanford: Design Thinking [www.youtube.com/watch?v=GeUXQ\\_L-35M&-feature=shares](http://www.youtube.com/watch?v=GeUXQ_L-35M&-feature=shares)
2. Design thinking workshop: [www.youtube.com/channel/UCOjS4V\\_nBkylZ-pnlJFMJnTw](http://www.youtube.com/channel/UCOjS4V_nBkylZ-pnlJFMJnTw)
3. Design Thinking HBR [www.youtube.com/watch?v=z3IbHLfcyWo](http://www.youtube.com/watch?v=z3IbHLfcyWo)
4. Design Thinking: Solving Life's Problems [www.youtube.com/watch?v=UQYoWwHg3qA](http://www.youtube.com/watch?v=UQYoWwHg3qA)

### Course Outcomes (COs):

On successful completion of the course, students will be able to

1. Understand design thinking (PO-2, PO-6)
2. Understand and initiate design thinking process (PO-2, PO-3, PO-6, PO-9, PO-10, PO-12)
3. Generate and develop design ideas and prototype (PO-2, PO-3, PO-6, PO-9, PO-10, PO-12)
4. Test and analyse failures (PO-2, PO-4, PO-6)
5. Learn how Design Thinking can be applied (PO-2, PO-3, PO-6, PO-9, PO-10, PO-12)

## Course Assessment and Evaluation:

| <b>Continuous Internal Evaluation (CIE): 50 Marks</b>                                                                         |              |                                 |
|-------------------------------------------------------------------------------------------------------------------------------|--------------|---------------------------------|
| <b>Assessment tool</b>                                                                                                        | <b>Marks</b> | <b>Course outcomes attained</b> |
| Internal Test-I                                                                                                               | 30           | CO1, CO2                        |
| Internal Test-II                                                                                                              | 30           | CO3, CO4, CO5                   |
| Average of the two internal test shall be taken for 30 marks                                                                  |              |                                 |
| <b>Other components</b>                                                                                                       |              |                                 |
| Assignment<br>MCQ<br>Quiz<br>Presentation<br>Model/Mini project<br>Any other (Casestudy/group activity<br>with report writing | 20           | CO1, CO2, CO3, CO4, CO5         |
| <b>Semester End Examination (SEE)</b><br>(Scaled to 50 Marks)                                                                 | 100          | CO1, CO2, CO3, CO4, CO5         |

---

## COMPUTER AIDED ENGINEERING DRAWING

---

**Course Code: MELV18**

**Credits: 0:1:1**

**Pre-requisites: -**

**Contact Hours: 14T+14P**

**Course Coordinator: Dr. Mohandas K N**

---

### Course Contents

#### Course Learning Objectives:

1. To understand the basic and applied engineering drawings solutions by using suitable software.
2. To comprehend the criticality of development of typical lateral surfaces.
3. To visualize typical engineering components.

#### Teaching-Learning (General Instructions):

- Students should be made aware of powerful engineering communication tool – Drawing.
- Simple Case studies can be suitably selected by the teacher for hands on practice to induce the feel of fruitfulness of learning.
- Appropriate Models, Power Point presentation, Charts, Videos, shall be used to enhance visualization before hands on practice.
- For application problems use very generally available actual objects. (Example: For rectangular prism / object; matchbox, carton boxes, book, etc can be used. Similarly for other shapes).
- Use any CAD software for generating orthographic and pictorial views.
- Make use of sketch book with graph sheets for manual / preparatory sketching.

### Unit I

#### Introduction:

Significance of Engineering drawing, BIS Conventions of Engineering Drawing, Free hand sketching of engineering drawing, Scales. Introduction to Computer Aided Drafting software, Co-ordinate system and reference planes HP, VP, RPP & LPP of 2D environment. Commands and creation of Lines, coordinate points, axes, polylines, square, rectangle, polygons, splines, circles, ellipse, text, move, copy, off-set, mirror, rotate, trim, extend, break, chamfer, fillet and curves.

#### Orthographic Projections of Points, Lines and Planes:

Orthographic projections: Definitions - Planes of projection, reference line and conventions employed, Projections of points in First and Third quadrants (**No problems**), Projections of straight lines (located in First quadrant/first angle only), True and apparent lengths, True and apparent inclinations to reference planes (**No problems**).



Orthographic Projections of Planes viz triangle, square, rectangle, pentagon, hexagon, and circular laminae (**Placed in First quadrant only using change of position method**).

- Pedagogy/Course delivery tools: Chalk and talk, Power point presentation
- Links: Orthographic Projections: <https://youtu.be/uePTMVQIIA4>
- Links: Orthographic projections of plane: <https://www.youtube.com/watch?v=GguSUMNxc8Q>

## Unit II

### Development of Lateral Surfaces of Solids:

Development of sectioned (**Section plane perpendicular to VP and inclined to HP bisecting the axis only**) lateral surfaces of right regular prism, cylinder, pyramid and cone resting **with base on HP only**.

- Pedagogy/Course delivery tools: Chalk and talk, Power point presentation
- Links: Development of lateral surfaces: [https://www.youtube.com/watch?v=U5mz9\\_W-xdI](https://www.youtube.com/watch?v=U5mz9_W-xdI)

## Unit III

### Orthographic Projection of Solids:

Orthographic projection of right regular solids (**Solids Resting on HP only**): Prism & Pyramid (triangle, square, rectangle, pentagon, hexagon), Cylinder, Cone and Cube (**No freely suspended problems**).

- Pedagogy/Course delivery tools: Chalk and talk, Power point presentation
- Links: Orthographic Projection of Solids: <https://www.youtube.com/watch?v=HiXD2qIqoGE&t=16s>

## Unit IV

### Isometric Projections:

Isometric scale, Isometric projection of hexahedron (cube), right regular prism, pyramid, cylinder, cone, sphere and frustum of solid. Isometric projection of combination of two simple solids (**Co-axial only**).

- Pedagogy/Course delivery tools: Chalk and talk, Power point presentation
- Links: Isometric Projection: [https://www.youtube.com/watch?v=Vo9LC9d7FQA&list=PLIhUrsYr8yHxVky7bfrnbRcdXcHjT\\_K83](https://www.youtube.com/watch?v=Vo9LC9d7FQA&list=PLIhUrsYr8yHxVky7bfrnbRcdXcHjT_K83)

## Unit V

### Multidisciplinary Applications & Practice: (For CIE only)

**Free hand Sketching:** True free hand, Guided Free hand, Roads.

**Drawing Simple Mechanisms:** Bicycles, Tricycles, Gear trains.

**Electric Wiring and lighting diagrams:** Like, Automatic fire alarm using suitable software

**Basic Building Drawing:** Architectural floor plan, basic foundation drawing using suitable software.

**Electronics Engineering Drawings:** Simple Electronics Circuit Drawings, practice on layers concept

- Pedagogy/Course delivery tools: Chalk and talk, Power point presentation
- Links: Free hand sketching: <https://www.youtube.com/watch?v=Ess0dmJB2lo>
- Links: Electric wiring and lighting diagram: <https://www.youtube.com/watch?v=c67wIH2IJL8>
- Links: Electronics Engineering Drawings: <https://www.youtube.com/watch?v=RpF7oFC-LPY>

### Course Outcomes (COs):

At the end of the course the student will be able to:

1. Demonstrate the usage of suitable software for creating basic and applied engineering drawings. (PO-1, PO-5, PO-9, PO-10, PO-12)
2. Conceptually sketch and draw developments for typical lateral surfaces. (PO-1, PO-5, PO-9, PO-10)
3. Exhibit the knowledge of orthographic and isometric projections of typical solids. (PO-1, PO-5, PO-9, PO-10)

### Course Assessment and Evaluation:

| <b>Continuous Internal Evaluation (CIE) : 50</b>  |              |                                  |
|---------------------------------------------------|--------------|----------------------------------|
| <b>Assessment tool</b>                            | <b>Marks</b> | <b>Course outcomes addressed</b> |
| Class work<br>(Drawing printouts and sketch work) | 30           | CO1, CO2, CO3                    |
| Class-room open book assignments                  | 10           | CO1, CO2, CO3                    |
| Internal assessment tests & surprise tests        | 10           | CO1, CO2, CO3                    |
| <b>Semester End Examination (SEE)</b>             | <b>50</b>    | <b>CO1, CO2, CO3</b>             |

---

## ENGINEERING CHEMISTRY LAB

---

**Course Code: CYLC29**

**Credits: 0:0:1**

**Pre-requisites: -**

**Contact Hours: 14P**

**Course Coordinators: Dr. Nagaraju Kottam & Dr. M N Manjunatha**

---

### Course Contents

#### **PART A: Any Five (choice based) - Instrumental**

1. Estimation of copper present in electroplating effluent by optical sensor (Colorimetrically).
2. Estimation of acid mixture (HCl & CH<sub>3</sub>COOH) by conductometric method.
3. Analysis of iron content present in E-waste effluent by potentiometrically.
4. Determination of pKa of vinegar using pH sensor (Membrane electrode - Glass electrode).
5. Estimation of sodium present in soil/an effluent sample using flame photometer.
6. Determination of Viscosity Coefficient of a lubricant sample (Ostwald's Viscometer).

#### **PART B: Any Five (choice-based)-Volumetric and other Techniques**

1. Assessment of suitability of drinking and industrial water by estimation of total hardness by EDTA method.
2. Determination of COD of an industrial effluent sample.
3. Estimation of percentage of iron in rust solution.
4. Estimation of copper in electroplating effluent sample iodometrically.
5. Synthesis of semiconducting metal oxide (ZnO/Fe<sub>2</sub>O<sub>3</sub>/CuO/Al<sub>2</sub>O<sub>3</sub>) nanomaterial.
6. Determination of rate of corrosion of mild steel by weight loss method.

#### **PART C: (Demo Experiments)**

1. Verification of Nernst's equation and measurement of single electrode potential.
2. Chemical structure drawing using software: Origin/ ACD/ChemSketch.
3. Analysis of cement by volumetric method
4. Synthesis of Urea-Formaldehyde polymer.
5. Preparation of a conducting polymer.

## Suggested Learning Resources:

### Reference books:

1. Vogel's quantitative Chemical Analysis, Pearson Publication, 6<sup>th</sup> edition, 2009.
2. **Clair N. Sawyer and Perry L, Mc. Carty** - Chemistry for Environmental Engineering, Mc. Graw-Hill Book Company, New York, 5<sup>th</sup> edition, 2003.
3. Chemistry Manual – RIT, 2024-25

### Web links and Video Lectures (e-Resources):

1. <http://msrit.edu/study-material/chemistry.html>

### Course Outcomes (COs):

At the end of the course the student will be able to:

1. Handle the apparatus in chemical laboratories for analysis of various materials (PO-1, PO-2, PO-7).
2. Analyze the suitability of water for domestic and industrial consumption (PO-1, PO-2, PO-7).
3. Synthesis of nano structured semiconducting and polymeric materials and evaluate the content and composition of new materials encountered in engineering applications (PO-1, PO-2, PO-7).

### Course Assessment and Evaluation:

| <b>Continuous Internal Evaluation (CIE): 50 Marks</b>                                                                                                |              |                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----------------------------------|
| <b>Assessment tool</b>                                                                                                                               | <b>Marks</b> | <b>Course outcomes addressed</b> |
| Weekly evaluation of laboratory manuals/records after the Conduction of every experiment.10 Marks x 10 experiments<br>100 Marks, reduced to 30 Marks | 30           | CO1, CO2, CO3                    |
| Weekly evaluation of laboratory manuals/records after the Conduction of every experiment.10 Marks x 10 experiments<br>100 Marks, reduced to 30 Marks | 20           | CO1, CO2, CO3                    |
| <b>Semester End Examination (SEE)</b>                                                                                                                | 50           | CO1, CO2, CO3                    |

---

## RENEWABLE ENERGY SOURCES

---

**Course Code:** : ETC144/244

**Credits:** 3:0:0

**Pre-requisites:** Nil

**Contact Hours:** 42L

**Course Coordinator:** Gururaj

---

### Course Content

The Students shall:

1. To understand energy layout, energy sources and their utilization.
2. To explore society's present needs and future energy demands.
3. To Study the principles of renewable energy conversion systems.
4. To exposed to energy conservation methods.

### Unit I

**Introduction:** Principles of renewable energy; energy and sustainable development, fundamentals and social implications, renewable energy availability in India, applications of renewable energy, Introduction to Internet of energy (IOE).

- Pedagogy/Course delivery tools: Chalk and talk, Power point presentation, animated videos
- Lab component/Practical topics: NCES lab
- Links: Introduction of renewable energy: <https://www.youtube.com/watch?v=A2KvSLoonGs>
- Links: Introduction to Internet of energy (IOE): <https://www.youtube.com/watch?v=jnf-aM197I0>

### Unit II

**Solar Energy:** Fundamentals; Solar Radiation; Estimation of solar radiation on horizontal and inclined surfaces; Solar radiation Measurements- Pyrheliometers, Pyrometer, Sunshine Recorder. Solar Thermal systems: Flat plate collector; Solar distillation; Solar pond electric power plant.

**Solar electric power generation-** Principle of Solar cell, Photovoltaic system for electric power generation, advantages, Disadvantages and applications of solar photovoltaic system.

- Pedagogy/ Course delivery tools: Chalk and talk, Power point presentation
- Lab component/Practical topics: NCES lab
- Links: Solar radiation Measurements: <https://www.youtube.com/watch?v=PmkbJx1jdV4>
- Links: Solar cell: <https://www.youtube.com/watch?v=mCgXsEyQZSI>

### Unit-III

**Wind Energy:** Properties of wind, availability of wind energy in India, wind velocity and power from wind; major problems associated with wind power, Basic components of wind energy conversion system (WECS); Classification of WECS- Horizontal axis-single, double and muliblade system. Vertical axis- Savonius and darrieus types.

**Biomass Energy:** Introduction; Photosynthesis Process; Biofuels; Biomass Resources;

Biomass conversion technologies -fixed dome; Urban waste to energy conversion; Biomass gasification (Downdraft).

- Pedagogy/ Course delivery tools: Chalk and talk, Power point presentation, animated videos,
- Lab component/Practical topics: NCES lab
- Links: Wind energy: <https://www.youtube.com/watch?v=f0p0Fria5TY>
- Links: Biomass energy: [https://www.youtube.com/watch?v=5cQD\\_xZph9U](https://www.youtube.com/watch?v=5cQD_xZph9U)

## Unit IV

**Tidal Power:** Tides and waves as energy suppliers and their mechanics; fundamental characteristics of tidal power, harnessing tidal energy, advantages and limitations.

**Ocean Thermal Energy Conversion:** Principle of working, OTEC power stations in the world, problems associated with OTEC.

**Geothermal Energy Conversion:** Principle of working, Geothermal sources, problems associated with geothermal energy conversion, advantages and limitations.

- Pedagogy/ Course delivery tools: Chalk and talk, Power point presentation, animated videos
- Lab component/Practical topics: NCES lab
- Links: Tidal energy: <https://www.youtube.com/watch?v=pnK5rv5PoSU>
- Links: OTEC energy: <https://www.youtube.com/watch?v=ala3ruvZMho>

## Unit V

**Green Energy:** Introduction, Fuel cells: Classification of fuel cells – H<sub>2</sub>; Operating principles, Zero energy Concepts. Benefits of hydrogen energy, hydrogen production technologies (electrolysis method only), hydrogen energy storage, applications of hydrogen energy, problem associated with hydrogen energy.

- Pedagogy/ Course delivery tools: Chalk and talk, Power point presentation,
- Lab component/Practical topics: NCES lab
- Links: Fuel cells: <https://www.youtube.com/watch?v=6oeN9VDFLig&t=3s>
- Links: Hydrogen energy: <https://www.youtube.com/watch?v=LdRelOJKMCs>

### Text Books:

1. **G D Rai** - Nonconventional Energy sources, Khanna Publication, 4<sup>th</sup> edition.
2. **S.Rao and Dr. B.B. Parulekar** - Energy Technology, Khanna Publication.

### Reference Books:

1. **A. W. Culp Jr.** - Principles of Energy conversion, McGraw Hill, 1996
2. **Shobh Nath Singh** - Non-Convention Energy Resources, Pearson, 2018

### Web links and video lectures (e-Resources)

- <https://www.pdfdrive.com/non-conventional-energy-sources-e10086374.html>
- <https://www.pdfdrive.com/non-conventional-energy-systems-nptel-d17376903.html>
- <https://www.pdfdrive.com/renewable-energy-sources-and-their-applications-e33423592.html>

- <https://www.pdfdrive.com/lecture-notes-on-renewable-energy-sources-e34339149.html>
- <https://onlinecourses.nptel.ac.in/noc18ge09/preview>

## Course Outcomes (COs):

At the end of the course, the student will be able to

1. Identify the various sources of renewable energy sources and their applications. (PO-1, PO-3, PO-5, PO-6, PO-7, PO-9, PO-11, PO-12)
2. Understand the working principles of solar energy extraction and its measurements. (PO-1, PO-3, PO-5, PO-6, PO-7, PO-9, PO-11, PO-12)
3. Describe the classifications, energy conversion and problems associated with wind energy and Biomass energy. (PO-1, PO-3, PO-5, PO-6, PO-7, PO-9, PO-11, PO-12)
4. Recognize the energy harvesting of Tidal energy, OTEC and geothermal energy. (PO-1, PO-3, PO-5, PO-6, PO-7, PO-9, PO-11, PO-12)
5. Understand the operation of fuel cell and hydrogen energy extraction. (PO-1, PO-3, PO-5, PO-6, PO-7, PO-9, PO-11, PO-12)

## Course Assessment and Evaluation:

| <b>Continuous Internal Evaluation (CIE): 50 Marks</b>          |              |                                 |
|----------------------------------------------------------------|--------------|---------------------------------|
| <b>Assessment Tool</b>                                         | <b>Marks</b> | <b>Course outcomes attained</b> |
| Internal Test-I                                                | 30           | CO1, CO2, CO3                   |
| Internal Test-II                                               | 30           | CO3, CO4, CO5                   |
| Average of the two internal tests shall be taken for 30 marks. |              |                                 |
| <b>Other Components</b>                                        |              |                                 |
| Assignment/Mini Project                                        | 10           | CO1, CO2, CO3, CO4, CO5         |
| Quiz/Presentations / Assignments                               | 10           | CO1, CO2, CO3, CO4, CO5         |
| <b>Semester End Examination (SEE)</b><br>(Scaled to 50 Marks)  | 100          | CO1, CO2, CO3, CO4, CO5         |

---

## INTRODUCTION TO CIVIL ENGINEERING

---

**Course Code:** ESC131/231

**Credits:** 3:0:0

**Pre-requisites:** Nil

**Contact Hours:** 42L

**Course Coordinator:** Dr. Basavangowda G M

---

### Course Content

#### Unit I

##### **Civil Engineering Disciplines and Building Science**

**Introduction to Civil Engineering:** Surveying, Structural Engineering, Geotechnical Engineering, Hydraulics & Water Resources, Transportation Engineering, Environmental Engineering, Construction planning & Project management.

**Basic Materials of Construction:** Bricks, Cement & mortars, Plain, Reinforced & Pre-stressed Concrete, Structural steel, Construction Chemicals.

**Structural elements of a building:** Foundation, plinth, Masonry wall, column, beam, slab, and staircase.

- Pedagogy/Course delivery tools: Chalk and Talk, Power point presentations, Videos
- Links: Introduction: <https://a.impartus.com/ilc/#/video/id/534326>
- Scopes of Various fields: <https://a.impartus.com/ilc/#/video/id/536441>

**Self Study:** Building components such as Lintel and Chejja

#### Unit II

##### **Societal and Global Impact of Infrastructure**

**Infrastructure:** Introduction to sustainable development goals, Smart city concept, Safe city concept

**Environment:** Water Supply and Sanitary systems, urban air pollution management, Solid waste management, urban flood control

**Built-environment:** Energy efficient buildings, recycling, Temperature and Sound control in buildings, Security systems; Smart buildings.

- Pedagogy/ Course delivery tools: Chalk and Talk, Power point presentations, Videos
- Links: Infrastructure Habitats: <https://www.youtube.com/watch?v=wpvbVyUCi78>  
<https://www.youtube.com/watch?v=Irq4BigInDU>

**Self Study:** Clean city concept, identification of Landfill sites.

#### Unit III

**Analysis of force systems:** Concept of idealization, system of forces, principles of superposition and transmissibility, Resolution and composition of forces, Resultant of concurrent and non-concurrent coplanar force systems, moment of forces, couple, Varignon's theorem, free body diagram, equations of equilibrium, equilibrium of concurrent and non-concurrent coplanar force systems.



- Pedagogy/Course delivery tools: Chalk and Talk, Power point presentations, Videos
- Links: Introduction to:  
Engg. Mechanics - <https://a.impartus.com/ilc/#/video/id/532408>  
Resultants - <https://a.impartus.com/ilc/#/video/id/537603>  
Equilibrium Concepts: <https://a.impartus.com/ilc/#/video/id/550330>

**Self Study:** Law of Parallelogram of forces, Numerical on Couple, and moment.

## Unit IV

**Centroid:** Importance of centroid and centre of gravity, methods of determining the centroid, locating the centroid of plane laminae from first principles, centroid of built-up sections. Numerical examples.

**Friction:** Definition of friction and its application, angle of friction, angle of repose, coefficient of friction, Types of friction, laws of static friction, Description, and application of friction on blocks on horizontal and inclined planes.

- Pedagogy/Course delivery tools: Chalk and Talk, Power point presentations, Videos
- Links: [https://www.youtube.com/watch?v=QK\\_TuE2IfSc](https://www.youtube.com/watch?v=QK_TuE2IfSc)
- Friction concepts: <https://www.youtube.com/watch?v=AlenFWwK3Ek>

**Self Study:** Derivation of Centroid of semicircle and quarter circle.

## Unit V

**Moment of inertia:** Importance of Moment of Inertia, method of determining the second moment of area (moment of inertia) of plane sections from first principles, parallel axis theorem and perpendicular axis theorem, section modulus, radius of gyration, moment of inertia of built-up sections, Numerical Examples.

- Pedagogy/Course delivery tools: Chalk and Talk, Power point presentations
- Links: Moment of Inertia: <https://www.youtube.com/watch?v=Bls5KnQOWkY>

**Self Study:** Derivation of moment of inertia of semicircle and quarter circle.

### Text Books:

1. **Bansal R. K., Rakesh Ranjan Beohar and Ahmad Ali Khan** - Basic Civil Engineering and Engineering Mechanics, Laxmi Publications, 2015.
2. **Kolhapure B K** - Elements of Civil Engineering and Engineering Mechanics, EBPB, 2014.

### Reference Books:

1. **Beer F.P. and Johnston E. R.** - Mechanics for Engineers, Statics and Dynamics, McGraw Hill, 1987.
2. **Irving H. Shames** - Engineering Mechanics, Prentice-Hall, 2019.
3. **Hibbler R. C.** - Engineering Mechanics: Principles of Statics and Dynamics, Pearson Press, 2017.
4. **Timoshenko S, Young D. H., Rao J. V.** - Engineering Mechanics, Pearson Press, 5<sup>th</sup> edition, 2017.

5. **Bhavikatti S S** - Engineering Mechanics, New Age International, 2019.
6. **Reddy Vijaykumar K and Suresh Kumar K**, Engineering Mechanics, BS publication, 2011.

### Web links and video Lectures (e- Resources):

1. <https://www.youtube.com/watch?v=nGfVTNfNwnk&list=PLOSWwFV98rfKXq2KBphJz95rao7q8PpwT>
2. <https://www.youtube.com/watch?v=nkg7VNW9UCc&list=PLOSWwFV98rfKXq2KBphJz95rao7q8PpwT&index=2>
3. <https://www.youtube.com/watch?v=3YBXteL-qY4>
4. <https://www.youtube.com/watch?v=z95UW4wwzSc&list=PLOSWwFV98rfKXq2KBphJz95rao7q8PpwT&index=10>
5. <https://www.youtube.com/watch?v=lheoBL2QaqU&list=PLOSWwFV98rfKXq2KBphJz95rao7q8PpwT&index=7>
6. [https://www.youtube.com/watch?v=atoP5\\_DeTPE](https://www.youtube.com/watch?v=atoP5_DeTPE)
7. <https://www.youtube.com/watch?v=kmsmp9OzAsI>
8. <https://www.youtube.com/watch?v=x1ef048b3CE>

### Course Outcomes (COs):

On successful completion of the course students will be able to

1. Understand the vast interface of civil engineering fields towards the society at large. (PO-1, PO-6)
2. Recognize the societal and global impact towards employment creation and its contribution to the GDP. (PO-1, PO-6, PO-7)
3. Analyze and evaluate the coplanar concurrent and non concurrent force system. (PO-1, PO-2, PO-3)
4. Locate the centroid of bodies and frictional force developed between the bodies. (PO-1, PO-2, PO-3)
5. Compute the moment of inertia of plane areas about a given axis. (PO-1, PO-2, PO-3)

### Course Assessment and Evaluation:

| <b>Continuous Internal Evaluation (CIE): 50 Marks</b>          |              |                                 |
|----------------------------------------------------------------|--------------|---------------------------------|
| <b>Assessment Tool</b>                                         | <b>Marks</b> | <b>Course outcomes attained</b> |
| Internal Assessment-I                                          | 30           | CO1, CO3, CO4                   |
| Internal Assessment-II                                         | 30           | CO2, CO4, CO5                   |
| Average of the two internal tests shall be taken for 30 marks. |              |                                 |
| <b>Other Components</b>                                        |              |                                 |
| Assignment                                                     | 10           | CO1, CO2                        |
| Assignment                                                     | 10           | CO3, CO4, CO5                   |
| <b>Semester End Examination (SEE)</b><br>(Scaled to 50 Marks)  | 100          | CO1, CO2, CO3, CO4, CO5         |